



Violence recognition on videos using two-stream 3D CNN with custom spatiotemporal crop

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Abstract

Violence may happen anywhere. One of the ways to know and oversee the violence in some places is by installing Closed-circuit Television (CCTV). The recorded video captured by CCTV can be used as proof in a law court. Violence video classification is also one of the topics being discussed in deep learning. The latest violence video dataset is RWF-2000. That dataset contains violent and non-violent videos, 5 seconds duration, 30 frames per second, with the amount of 2000 videos. That publication also has the best accuracy of 87.25% by their proposed method. In this study, we will use a Residual Network known to have the advantage of solving the vanishing gradient problem. Beside that, we also implement transfer learning from Kinetics and Kinetics + Moments in Time pre-trained data. We also test the number of frames and the location range of the sampled frames. RGB and optical flow inputs are separately trained with different configurations. The RGB input best accuracy is 89.25% with pre-trained Kinetics + Moments in Time, using frame location of 49–149. The optical flow input best accuracy is 88.5% with pre-trained Kinetics, using 74 frames. We also try to sum the output of both inputs making accuracy of 90.5%.

Keywords Video classification · ResNet · Transfer learning · Freeze layer

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1 Introduction

Violence may happen anywhere. One of the ways to oversee the violent activities in some places is by installing Closed-circuit Television (CCTV). CCTV is installed at places that should be supervised or potential crime areas like public places, the front of buildings, warehouses, and other important places. That also helped the security staff to know what happened at the same time in different places. Moreover, the recorded video captured by CCTV can be used as proof in law court [13].

Some cases that have been revealed due to CCTV footage in Indonesia are as follows: a man who beat his girlfriend in Cimahi [6], a nanny that abused two children [15], and a group of youth doing persecution [7]. Those cases can be adjudicated easier because of the evidence. The security system can also be maximized if there is a warning when the violence happens. That warning can be triggered as the result of violent video classification.

Video classification involves spatial (height and width of the frame) and temporal (frame changes per time). Its inputs are also based on color space (RGB) and optical flow. Video classification can use hand-crafted features using machine learning or a deep learning approach. There are some violence video datasets, such as Crowd Violence dataset [11], UCF-Crime dataset [25], and real-world fighting 2000 (RWF-2000) [5]. The use of feature fusion enhances the recognition performance due to its mutual information. It can be performed by aggregation, concatenation or utilizing additional fusion network. Research like [31] conducted fusion of two-stream CNN for action recognition. Furthermore, gating network is applied on fusion for dynamic stream selection. In medical imaging, fusion mechanism is conducted to produce precise classification [1] and segmentation [2].

In the Crowd Violence dataset publication, the highest evaluation value is 81.30% accuracy, reached by Violent Flows (ViF) with the machine learning approach. It contains 246 videos with a duration of 1.04–6.52 seconds about the riot in the crowd. UCF-Crime has about 1900 videos with 60–600 seconds of recorded surveillance camera representing unusual events such as accidents, explosions, violence, and theft. That publication has the best AUC of 75.41 as the best evaluation value by the proposed method with constraint. The latest dataset focusing on violent videos is RWF 2000, which includes 2000 videos labeled as fight or non-fight. Flow gated networks achieved the highest accuracy of 87.25% with 270 thousand parameters.

In research about video classification, Residual Network-based model (ResNet) shows better performance compared to VGG-16 on UCF-101 and HMDB-51 datasets with minimum parameters [31]. ResNet as a backbone has good performance especially ResNet-18 with RGB inputs [29]. The performance of ResNet-18 for video classification can also be improved by data augmentation using spatial crop [10].

Thus, based on aforementioned reviews, the contributions of the paper can be summarized as follows:

- A comprehensive review of the existing works in video based violence classification
- Propose two-stream 3D CNN with custom spatiotemporal crop for video based violence classification
- Evaluation of the performance of the proposed two-stream 3D CNN with custom spatiotemporal crop through experiments and comparison with the state-of-the-art methods on large scale RWF-2000 dataset

The organization of this paper is as follows. In Section 2, we present the related works on video and violence classification and its state-of-the-arts problems and models. In Section 3,

we describe the dataset and proposed models used in our experiments. In Section 4, we present the experimental setup, including pre-processing, training and testing methodology, and fusion of RGB and optical flow. In Section 5, we evaluate and discuss the performance of the proposed method through experiments and comparisons with the state-of-the-art methods. Finally, in Section 6, we conclude the paper and suggest future work.

2 Related works

Many approaches like machine learning have done violence video classification in Crowd Violence Dataset and Deep learning in UCF-Crime and RWF-2000. The latest datasets, RWF-2000, have been used as a benchmark in some latest research. For example, in RWF-2000 publication [5], the best accuracy is 87.25% by Flow Gated Network with about 270 thousand parameters, followed by I3D (RGB only) [4] in the second place with 85.75% accuracy and 12.3 Millions of parameters. In [22], the authors proposed a two-stream CNN architecture that uses RGB and optical flow information to perform video classification. The authors reported an accuracy of 85.1% on the UCF-Crime dataset. In [18], a multi-scale of VGG-19 architecture was proposed for violence video classification. The authors reported an best accuracy of 90.47% on the RWF-2000 dataset, outperforming previous state-of-the-art methods. In [19], the authors proposed a novel attention mechanism that focuses on the spatiotemporal regions of the video that are most relevant for violence classification. The authors reported accuracy improvements on the four datasets, demonstrating the effectiveness of the proposed attention mechanism. In [9], a Gated Recurrent Unit (GRU) network was proposed for violence video classification. The authors reported an accuracy of 86.43% on the RWF-2000 dataset, demonstrating the effectiveness of the GRU network for violence video classification. However, gaining help from other modalities is required to advance the performance. In [28], the authors proposed a Siamese network architecture for one-shot violence video learning. The authors reported an accuracy of improvement on the UCF-Crime dataset, demonstrating the effectiveness of the Siamese network over I3D and C3D model. In [30], the authors proposed a two-stream C3D network architecture that uses RGB and optical flow information to perform violence video classification. The authors reported best accuracy of 96.5% on the Hockey dataset, demonstrating the effectiveness of the proposed two-stream CNN architecture over another deep models and machine learning. Another work [21] proposed a vision transformer network for violence video classification. The authors reported an best accuracy on the Hockey dataset over the CNN-based network, showing the effectiveness of the proposed approach. However, the high number of parameters revealing the need to optimize transformer-based network (Fig. 1).

In [16], the model used the attention module SAM (Spatial Attention Module) for data processing, ghost module, and LSTM (Long-short Term Memory) in the model. Moreover, the model implements maximum pooling, and average pooling, respectively. Later, the feature maps are concatenated (CAT). That form an efficient feature indicator. After concatenated, it performs convolution operations through the convolution layer to extract the critical information area of the feature map and suppress irrelevant information. As a result, the model produces 87.5% accuracy on RWF-2000. Violence detection with Separable Convolutional LSTM [12] gets the highest accuracy by 89.75% with around 333 thousand parameters. The stream inputs using background suppressed frames and frame differences. The model also

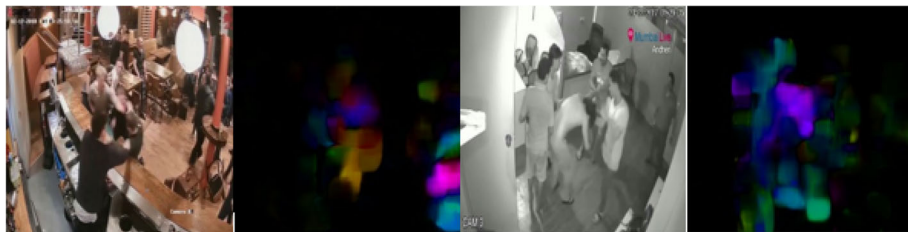


Fig. 1 Clips from RWF-2000

involves MobileNet as the CNN step. The fusion strategies used Leaky ReLU activation in feature maps of frames stream. They performed an element-wise multiplication to the sigmoid activation layer in the feature maps from the frame difference stream. Overall, these works demonstrate that deep learning methods can be effective for violence video classification, however, the choice of method will depend on the specific requirements of the task, such as computational resources, accuracy, and data availability.

In other research about video classification, Residual Network (ResNet) has the potential to get good classification performance like in [31] compared to VGG-16. The residual network also performs well as a backbone, especially ResNet-18 in [29]. The performance of the architecture can be increased by using data augmentation and transfer learning. As we can see in [10] and [14], ResNet-18 performance improved by using random spatial crop and transfer learning from pre-trained on Kinetics and Kinetics + Moments in Time.

3 Proposed methodology

3.1 Dataset

Some datasets contain violent acts like Crowd Violence, UCF-Crime, and RWF-2000. RWF-2000 is the latest dataset that contains violent acts and has a total of 2000 videos with two labels (fight and non-fight). The videos were gathered by scraping data from the Youtube website, specifically keywords related to violence with various resolution standards. The video content is captured from the perspective of surveillance cameras and involves the act of anomalous behavior in two, multiple people, and even a crowd. It also makes the videos close to realistic violent events. Each video has a duration of 5 seconds with 30 frames per second. RWF-2000 has a proportion of 80% of the training set and 20% of the testing set [5].

3.2 3D residual networks (3D ResNet)

3D Residual Networks is an improvement from the previous ResNet with adding the dimension of the input to store temporal information using 3D Convolution and 3D pooling. The tensor have 4 dimensions with size of $N_i \times L \times H_i \times W_i$ where N_i is the number filter used in block- i , L is the length of frame used as input, and $H_i \times W_i$ is the dimension of the frame. The number of frames as default input is 16 and the frame size is 112×112 . [27]. We can see the architecture of 3D ResNet-18 in Table 1.

3.3 Two-stream 3D residual networks (3D ResNet)

```

import torch
import torch.nn as nn
import torch.optim as optim

# Define the two 3D CNN models for the RGB and optical flow streams
rgb_stream = nn.Sequential(
    nn.Conv3d(3, 32, kernel_size=(3,3,3), stride=1, padding=1),
    nn.ReLU(),
    nn.MaxPool3d(kernel_size=(2,2,2), stride=2),
    nn.Flatten(),
    nn.Linear(32 * X * Y * Z, 128)
)

optical_flow_stream = nn.Sequential(
    nn.Conv3d(2, 32, kernel_size=(3,3,3), stride=1, padding=1),
    nn.ReLU(),
    nn.MaxPool3d(kernel_size=(2,2,2), stride=2),
    nn.Flatten(),
    nn.Linear(32 * X * Y * Z, 128)
)

# Define the optimizer and loss function for the models
optimizer = optim.SGD(list(rgb_stream.parameters()) +
list(optical_flow_stream.parameters()))
loss_function = nn.CrossEntropyLoss()

# Loop through each video in the training dataset
for video, labels in training_dataset:
    # Extract RGB frames and optical flow frames from the video
    rgb_frames = extract_rgb_frames(video)
    optical_flow_frames = extract_optical_flow_frames(video)

    # Convert the frames and labels to tensors
    rgb_frames = torch.tensor(rgb_frames)
    optical_flow_frames = torch.tensor(optical_flow_frames)
    labels = torch.tensor(labels)

    # Zero the gradients
    optimizer.zero_grad()

    # Compute the outputs from the two streams
    rgb_outputs = rgb_stream(rgb_frames)
    optical_flow_outputs = optical_flow_stream(optical_flow_frames)

    # Sum the two outputs
    outputs = rgb_outputs + optical_flow_outputs

    # Compute the loss
    loss = loss_function(outputs, labels)

    # Backpropagate the loss
    loss.backward()

    # Update the weights
    optimizer.step()

```

Listing 1 Pseudocode of Two-stream 3D CNN

The pseudocode 1 implements a training loop for a two-stream 3D CNN for video classification using PyTorch. The input to the network is RGB and optical flow frames of the video, and the output is the predicted class label for the video. The first part of the code imports the necessary libraries, including PyTorch and its neural network (nn) and optimization (optim) modules. Next, the two 3D CNN models for the RGB and optical flow streams are defined

Table 1 The architecture of 3D ResNet-18

Layer Name	3D ResNet-18
Conv1	$7 \times 7 \times 7$, 64, stride 1 (T), 2 (XY) $3 \times 3 \times 3$, max pool, stride 2
Conv2	$\begin{bmatrix} 3 \times 3 \times 3, 64 \\ 3 \times 3 \times 3, 64 \end{bmatrix} \times 2$
Conv3	$\begin{bmatrix} 3 \times 3 \times 3, 128 \\ 3 \times 3 \times 3, 128 \end{bmatrix} \times 2$
Conv4	$\begin{bmatrix} 3 \times 3 \times 3, 256 \\ 3 \times 3 \times 3, 256 \end{bmatrix} \times 2$
Conv5	$\begin{bmatrix} 3 \times 3 \times 3, 512 \\ 3 \times 3 \times 3, 512 \end{bmatrix} \times 2$
	Average Pool, C-d fc, softmax

using the `nn.Sequential` module. Each stream contains a series of convolutional (`nn.Conv3d`), activation (`nn.ReLU`), pooling (`nn.MaxPool3d`), and linear (`nn.Linear`) layers. Note that the input size for the optical flow stream is 2 instead of 3 as for the RGB stream because the optical flow is represented by two channels (horizontal and vertical optical flow). The optimizer is defined as the SGD optimizer and is initialized with the parameters of both the RGB and optical flow streams. The loss function used is the cross-entropy loss (`nn.CrossEntropyLoss`).

The training loop processes each video in the training dataset, extracting the RGB and optical flow frames from the video. The frames and labels are then converted to PyTorch tensors. The gradients are first zeroed in each loop iteration using `optimizer.zero_grad()`. The outputs from the RGB and optical flow streams are computed using the `rgb_stream` and `optical_flow_stream` models, respectively. The outputs are then summed to produce the final output. The loss is computed using the `loss_function`, predicted outputs, and true labels. The loss is then backpropagated using `loss.backward()`, and the weights are updated using `optimizer.step()`. This process continues until all the videos in the training dataset have been processed.

The two-stream 3D ResNet for violence video classification is a type of deep learning model used for the classification of violent behavior in videos as shown in Fig. 2. The two streams in the model represent two inputs to the network: RGB frames and optical flow (OF) frames. The 3D ResNet is a deep convolutional neural network architecture that is designed to handle the input of video data, which has a temporal dimension as well as spatial dimensions. The ResNet architecture incorporates residual connections, which allow the network to learn residual representations of the input data that can help to reduce overfitting and improve the performance of the model.

The RGB stream processes the RGB frames of the video and the OF stream processes the optical flow frames, which represent the movement of pixels in the video. The outputs from the two streams are then combined and fed into a final classifier layer, which outputs the predicted class label for the video. The two-stream 3D ResNet for violence video classification is trained on a large dataset of videos labeled as either violent or non-violent. During training, the model is optimized to minimize the classification loss between the predicted outputs and the true labels. After training, the two-stream 3D ResNet can be used for real-time violence video classification by processing the input RGB and OF frames through the two streams and combining the outputs to produce the final prediction.

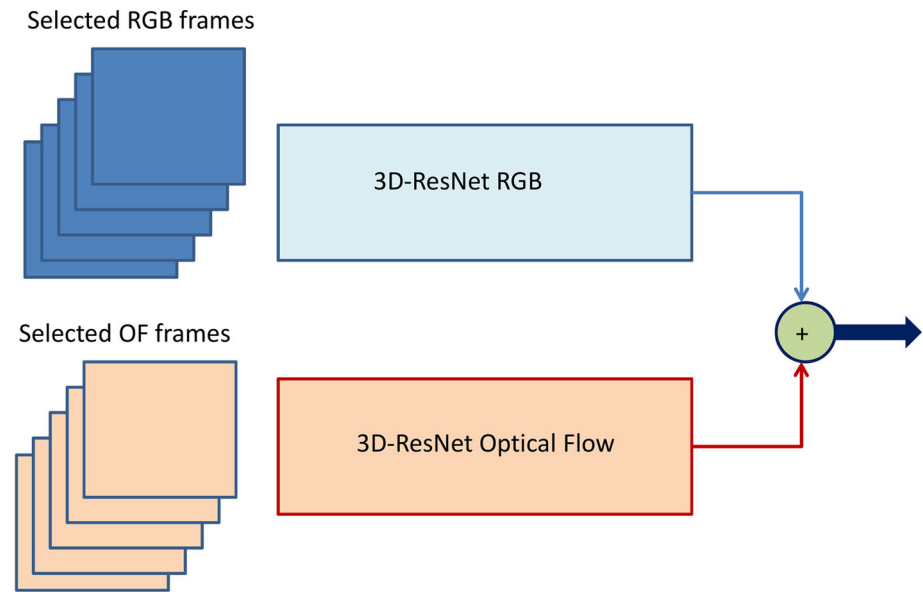


Fig. 2 Illustration of Two-stream 3D ResNet on Video classification

3.4 Data augmentation

3.4.1 Data augmentation: Random spatial crop

Random Spatial Crop will randomly crop the image with a certain ratio and scale. This augmentation will enhance the knowledge of the data. Note that it is applied during the training phase. The illustration of this augmentation can be seen in Fig. 3.

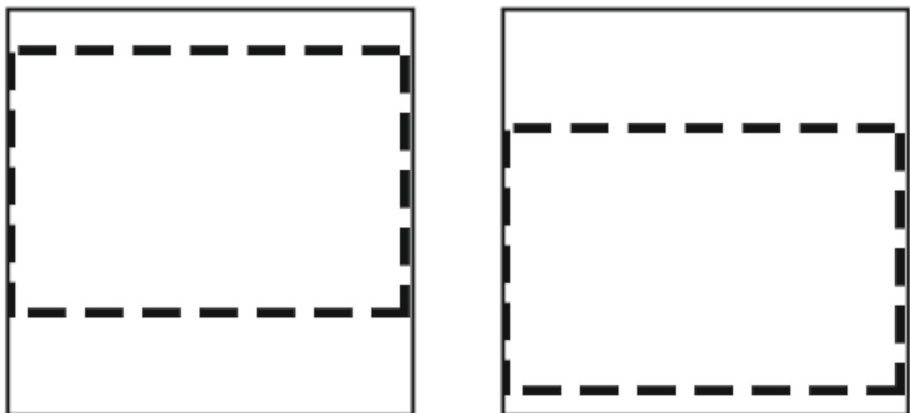


Fig. 3 Illustration of Random Spatial Crop



Fig. 4 Illustration of Random Spatiotemporal Crop

3.4.2 Data augmentation: Random spatiotemporal crop

Random Spatiotemporal Crop will crop the sequence of frames with location range of the sampled frames randomly. The maximum number of sampled frames will be initiated in advance. The cropped frame will still be in order. The box in Fig. 4 represents the frame. If the frame sampled at random is less than the number of the maximum number of frames, it will be looped. For example, the maximum number of frames is 5, but the sampling is 3: [67, 68, 69], thus it will be [67, 68, 69, 67, 68].

3.4.3 Data Augmentation: Video-wise Random Spatiotemporal Crop

Video-wise random spatiotemporal crop will randomly crop the frames in a certain number. For example, if the number of random frames is 50, it will choose 50 frames randomly from the full frames. Even though the frames are random, they will be sorted before the following processing. As the illustration in Fig. 5, the example of random frames are [a, b, c, d, e, f, g, h], but it will be [g, d, h, e, c, a, b, f] when it enters the model.

3.4.4 Data augmentation: Custom spatiotemporal crop

Custom Spatiotemporal Crop augments data by cropping the location range of the sampled frames based on specified start and endpoints. The start (a) and endpoint (b) must be initiated. The illustration of this augmentation can be seen in Fig. 6.

3.4.5 Data augmentation: Center spatial crop

Center Spatial Crop will crop the image in the center with a certain ratio and size. This will be used in testing for the consistency of data. Therefore, it will make the evaluation more reliable. The illustration of this augmentation can be seen in Fig. 7.

3.5 Transfer learning

Transfer Learning is a technique for fine-tuning a model already trained in different problems. Pre-trained models may increase the model's performance, mainly when we use a small dataset. The model that used a pre-trained model must have similar architecture. The study about transfer learning especially using 3D ResNet by [14] shows good performances. That publication also provides a pre-trained model for 3D ResNet-18 of Kinetics and Kinetics + Moments in Time dataset.

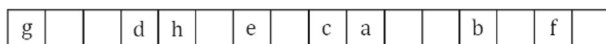


Fig. 5 Illustration of Video-wise Random Spatiotemporal Crop



Fig. 6 Illustration of Custom Spatiotemporal Crop

3.5.1 Kinetics

Kinetics is one of the datasets that have many classes about human action. In 2017, Kinetics had 400 classes with 306,245 videos. It was updated in 2018, becoming 600 classes and about 500,000 videos. Furthermore, the latest version of Kinetics-700 has 700 classes and around 650,000 videos. In addition, Kinetics-700 added more cooking, dancing, face-related action, body-only action, and more interactions with objects in video classes. The duration of the clips in datasets is 10 seconds. To verify the label, this dataset changes the scheme from using Mechanical Turk to a dedicated internal Google Team. The last filtering stage uses a crowd-sourced system [3].

3.5.2 Moments in time

Moments in Time dataset have three modalities: spatial, temporal, and auditory. It consists of 3-second corresponding, collected over 1,000,000 labeled videos for 339-moment classes. Over 1000 videos are associated with each verb. The dataset has various dynamic events involving different agents (people, animals, objects, and natural phenomena). The difficulty of this dataset is different levels of abstraction. For example, actions like opening can be

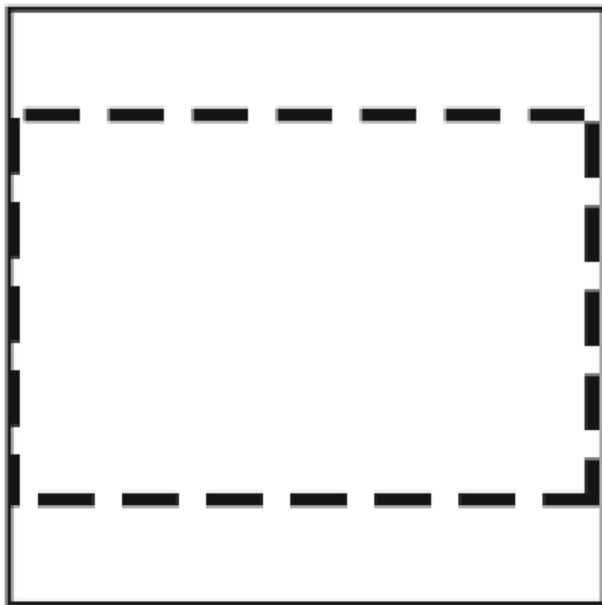


Fig. 7 Illustration of Center Spatial Crop

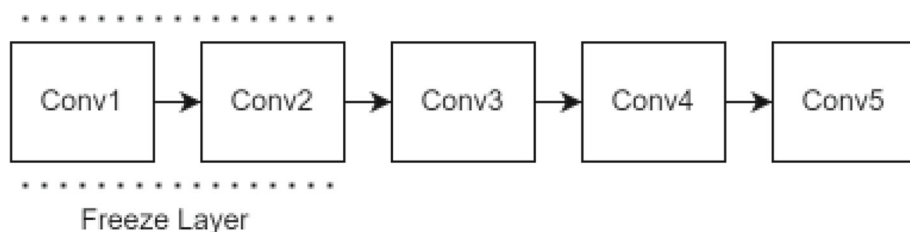


Fig. 8 Illustration of Freeze Layers

opening doors, gates, drawers, and opening eyes; however, if the set of frames in reverse depicts a different action (if the opening is reversed, it will act as closing) [17].

3.6 Freeze layer

Using a freeze layer will make some layers unable to be trained, or the parameter's value will be kept. The application of the freeze layer is to preserve the parameter value stored by the pre-trained model. The frozen layer can also accelerate training to make the model more effective and optimal [8]. The trained model will also produce a hierarchical representation that will transfer the input data as simpler features to the next layer (Fig. 8).

In this study, the freeze layer will only be used in the first and second layers. So that layers will not be updated, the parameters will remain as the first it initiated (in this case, using pre-trained). After that, the others will be trained and update the weight of the parameters.

4 Experimental setup

4.1 Pre-processing

This study extracts the video as an RGB image and optical flow. Extracting RGB images by captured frames per second generates 149 frames per video. We use Dense Optical Flow provided by OpenCV to get 74 optical flow images per video. The extracted image is resized by 224×224 as the standard of ImageNet.

4.2 Training methodology

We use some schemes for the training model. The input of RGB frame will be varied, starting with 16 as the default input of 3D ResNet, 30 as the frame per second in video, 50 as the $\frac{1}{3}$ total frames, 100 as the $\frac{2}{3}$ total frames and 149 as the full frames using random spatiotemporal crop. We are inspired by [20] to search the location range of the sampled frames. The location range of the sampled frames divided the total frames into three parts of 50, 100, and 149 frames. The location range of the sampled frames will be like 0-50, 0-100, 49-149, 50-100, 49-149, and 0-149. This scheme uses a custom spatiotemporal crop. We also train the model with video-wise random spatiotemporal crop using 50 and 100 frames as the $\frac{1}{3}$ and $\frac{2}{3}$ of full frames either from scratch or using transfer learning. The input of Optical Flow will only use one scheme of full frames (74 frames) with the same model. RGB and Optical Flow will be trained either from scratch or using transfer learning.

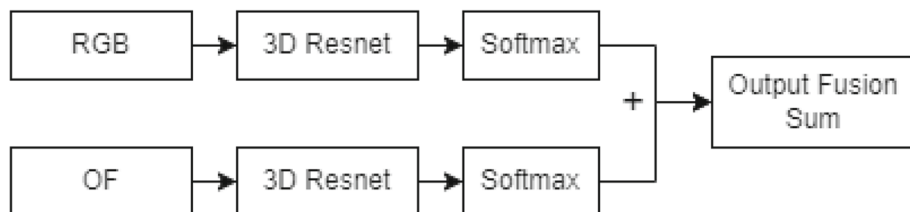


Fig. 9 Illustration of Fusion Sum of RGB + Optical Flow

The training model will use Stochastic Gradient Descent (SGD) as the optimizer with a learning rate scheduler. The learning rate scheduler divides the initial the learning rate by 10 for every 50 epochs, so in this study, the learning rate range is 0.01-0.0001. This study used an initial learning rate of 0.01 with a batch size of 32. All the training sessions will run for 200 epochs and checkpoint is done for every 20 epochs.

Before the inputs enter the model, all the data will be augmented. The default spatial input size is 112×112 , with the depth or temporal input varies depending on the testing scheme. The data augmentations that will be used in training are a random spatial crop for all cases, a random spatiotemporal crop for training with various RGB frame inputs, a custom spatiotemporal crop for training the location range of the sampled frames, and a video-wise random spatiotemporal crop. The minimum scale of the random spatial crop is 0.25, with the minimum ratio of 0.75.

4.3 Testing methodology

We use three schemes for testing the model, RGB only, Optical Flow only, and the fusion RGB + Optical Flow. The metric that will be used is accuracy. We use 400 videos with a full video frame (149 frames) as input. In this phase, there will be 40 batches with ten videos each. The performance of each training scheme will be evaluated here. Data augmentation testing is the center spatial crop. That makes data in testing tend to be normal. The size of the center crop is 112×112

4.3.1 Fusion RGB + Optical flow

The performance of video classification can be maximized by the fusion of the RGB and Optical Flow. In this study, we will sum the model's output and pooling to get the classification result. The best performance model of RGB and Optical Flow streams is selected as the final two-stream model. The illustration of Fusion RGB + Optical can be seen in Fig. 9.

5 Results and discussion

5.1 Number of frame input and random spatiotemporal crop

The first training scheme is to check the number of frames that perform the best accuracy using a random spatiotemporal crop from scratch. There is a high peak loss of 1.616 at epoch of 39 in validation. The graph in Fig. 10 also shows the lowest accuracy in the validation of 57.25% at that epoch. The unfit value of the parameter may cause that peak after the update

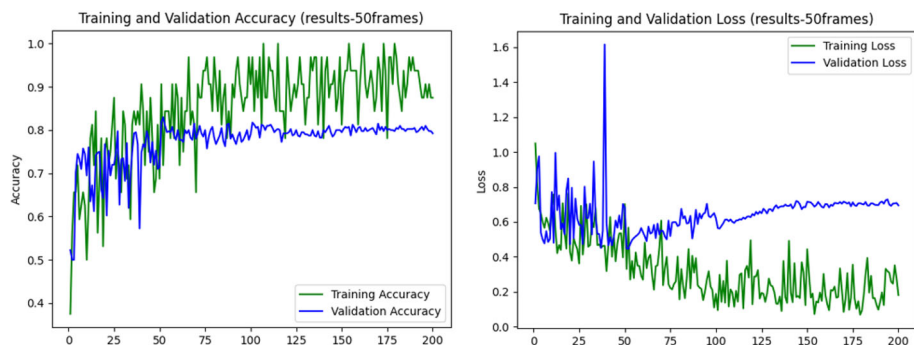


Fig. 10 The Graphs of Accuracy and Loss for The Best Performance Using Random Spatiotemporal Crop, maximum 50 frames from scratch

of its value. The update of its value is also affected by the learning rate. A high learning rate may change the value of parameters quite significantly; however it is hard to get convergence due to the changing exorbitant value. The learning rate decreases because of the multistep learning rate. Specifically, for every 50 epochs, learning rate is divided by 10, starting from 0.01 to 0.0001 as the maximum epoch is 200. It also affects the maximum peak per 50 epochs. The tendency of this graph scheme looks overfitting. The loss in training and validation has a gap. The accuracies in training hit 100%, however, the highest validation accuracy only reaches 83% at epoch of 52. The validation accuracy is consistently around 79.25–81.5% when the iteration is near the maximum epochs.

Table 2 shows the maximum accuracy of 83% by using a maximum of 50 frames as input, followed by a maximum of 100 frames inputs with 82% accuracy, and a maximum of 149 frames input with 81.75% accuracy. The default frames input, 16 frames, shows the worst performance in this scheme. The size of frame inputs affects the information that the model will retrieve. The lack of the number of frames used will produce minimum knowledge of the model. Otherwise, too many frames also make the information biased.

5.2 Location Range of Sampled Frames

The second training scheme uses a fixed location range of the sampled frames from scratch. This shows which part of video frame sampling has unique or feature values in this dataset. The frame location is strictly divided into three parts: beginning, middle, and end. There are 3 high peaks of loss in Fig. 11 when validation phase, respectively at epoch 7 (loss 1.324, accuracy 53%), epoch 18 (loss 1.284, accuracy 58.25%), and epoch 43 (loss 1.2627, accuracy 60.5%). That graph also shows the tendency of overfitting. There is still some loss gap between testing and validation. The anomaly occurs between epochs 60–85 when the

Table 2 Accuracy for Number of Frames as Input Using Random Spatiotemporal crop, Training from Scratch

Number of Frames	Train	Val
16	100	77.25
30	100	79.25
50	100	83
100	100	82
149	100	81.75

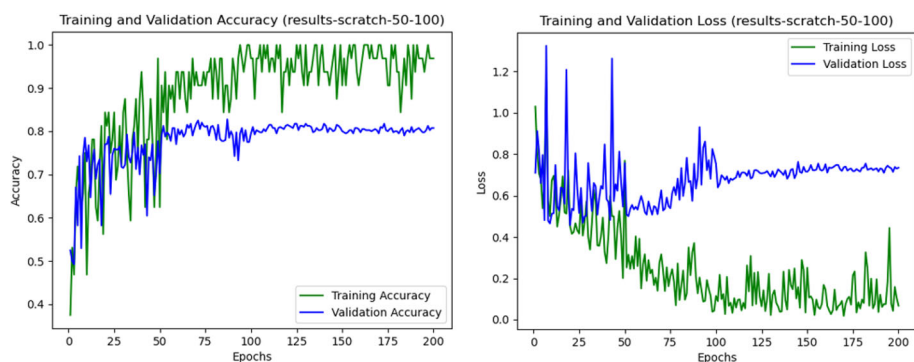


Fig. 11 The Graphs of Accuracy and Loss for The Best Performance Using Custom Spatiotemporal Crop, from scratch, Location Range of The Sampled Frames: 49-149

loss of training decreases but the loss of validation increases. It may be caused by the model that suits to the training dataset but not with the validation. Peak also happened at epoch of 195 of training loss, however, the performance in loss validation was still stable until the end of the epochs.

As shown from Table 3, the location range of the sampled frames of 50-100 has the best accuracy with 82.75%, followed by 49-149 with 82%. Middle to end frame location may show the location of the feature values. However, that does not prove the location of the violent act.

Video-wise random spatiotemporal augmentation performs with lower accuracy, as shown in Table 3. It may be caused by the loss of information given by the spatiotemporal. When using more frames to be randomized, the better its performance. Fewer randomized frames mean only a few insights from the spatiotemporal information. It also can skip the important frames that may be the keys to the information.

5.3 Using transfer learning and freeze layer

Transfer learning improves the performance significantly of the model 3D ResNet-18 in the RWF-2000 dataset. The available models are pre-trained from Kinetics and Kinetics +

Table 3 Accuracy for The Location Range of The Sampled Frames Using Custom Spatiotemporal Crop, Training from Scratch

The Location Range of The Sampled Frames	Train	Val
0-50	100	79.25
0-100	100	81.5
0-149	100	81.25
49-149	100	82
50-100	100	82.75
99-149	100	80.75
VWR-50	100	72.5
VWR-100	100	79.75

* VWR-50 and VWR-100 respectively stand for Videowise-Random-50 and Videowise-Random 100, using 50 and 100 frames with video-wise random spatiotemporal augmentation in training

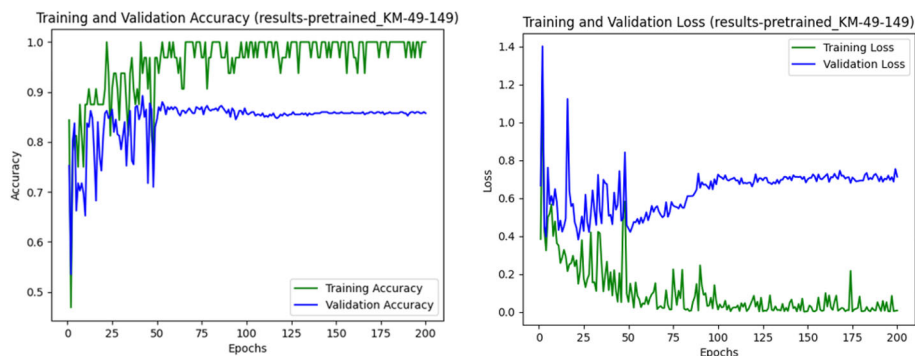


Fig. 12 The Graphs of Accuracy and Loss for The Best Performance Using Transfer Learning, Kinetics + Moments in Time Pre-trained, and Location Range of The Sampled Frames: 49-149

Moments in Time. The idea of Kinetics + Moments in Time pre-trained is training the model to the combined dataset Kinetics (700 classes) and Moments in Time (339 classes) with a total of 1039 classes. The model loss in Fig. 12 tends to show overfitting due to the gap of the loss between training and validation but still gets good accuracy, with maximum accuracy reached at epoch of 42 with the accuracy of 89.25 and loss of 0.6297. Although it overfits, the loss graph looks convergence both in training and validation. The graph also tells the first 50 epochs, model struggle to update its parameter, but it gets well as time goes by. Because for every 50 epochs, the learning rate is divided by 10, the learning rate decreases over time.

We can see in Table 4, The maximum accuracy was reached by using pre-trained from Kinetics + Moments in Time with 89.25% accuracy for the location range of the sampled frames of 49-149. This is because the implementation of transfer learning lets the model parameters already trained in the previous dataset be used and tuned into a new dataset. On the other side, using pre-trained Kinetics shows the same accuracy with 86.5%, however the

Table 4 Accuracy for Using Pre-trained Kinetics (K) and Kinetics + Moments in Time (K+M), Input RGB Frame

The Location of Sampling Frame Range	K		K+M	
	Train	Val	Train	Val
0-50	100	86	100	87
0-100	100	85	100	86.5
0-149	100	86.5	100	87.5
49-149	100	86.5	100	89.25
50-100	100	86.5	100	86.25
99-149	100	86.5	100	85.5
R-50	100	84.25	100	85.5
VWR-50	100	72.75	100	76.5
VWR-100	100	85.25	100	85.75

*R-50 stands for random-50, using 50 frames with random spatiotemporal augmentation in training

*VWR-50 and VWR-100 respectively stand for Videowise-Random-50 and Videowise-Random 100, using 50 and 100 frames with video-wise random spatiotemporal augmentation in training

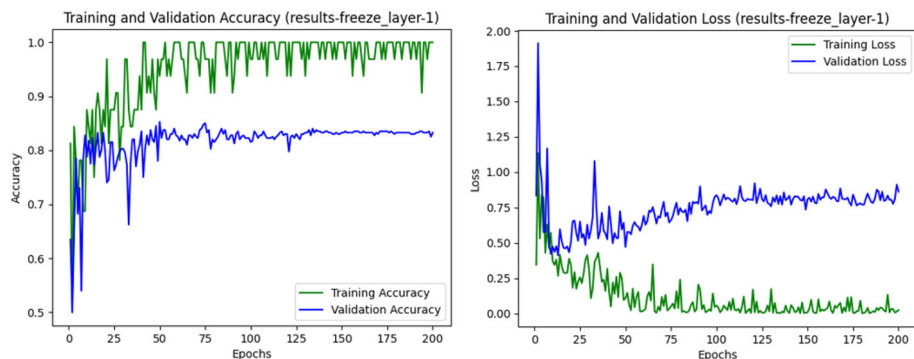


Fig. 13 The Graphs of Accuracy and Loss for Using 1 Freeze Layer, Pretrained Kinetics + Moments in Time Pre-trained, and Location Range of The Sampled Frames: 49-149

loss is different for the location range of the sampled frames of 0-149 with a loss of 0.449, frame range of 49-149 with a loss of 0.535, and frame range of 50-100 with loss of 0.5073.

Using random spatiotemporal augmentation, as shown in Table 4, the model significantly performs better than video-wise random spatiotemporal crop with 50 frames. Random spatiotemporal has the information of the successive frames because it only randomizes the initial or the starting position of the frame sampling. Video-wise random augmentation will choose a sorted random frame sampling. This can skip some pieces of information because it may skip the consecutive frame or also jump from the frame to frame too widely. The video-wise random spatiotemporal crop with 100 frames makes the model performs better than the random spatiotemporal crop with 50 frames. Using more frames will give more insight because of the more extended spatiotemporal information.

The RGB stream shows the best accuracy of 89.25 in Table 4 with pre-trained Kinetics + Moments in Time and location range of the sampled frames of 49-149. We also check that configuration by freezing the first and second layers. Note that the freeze layer will not update the value of parameters.

As we compare from Figs. 12, 13, and 14, the performance without freeze layer still have the best accuracy. The graphs look overfitting, and the gap of training and validation are also

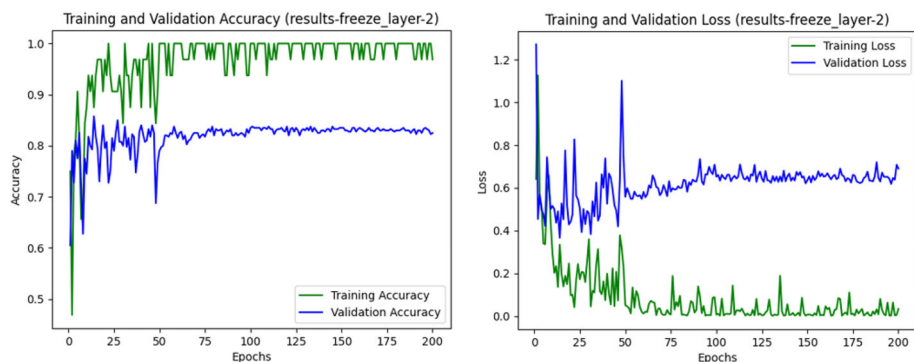


Fig. 14 The Graphs of Accuracy and Loss for Using 1 Freeze Layer, Kinetics + Moments in Time Pre-trained Model, and Location Range of The Sampled Frames: 49-149

Table 5 Accuracy of Various Freeze Layers

Number of Freeze Layer	Train	Val
0	100	89.25
1	100	85.25
2	100	85.75

evident. Using two freeze layers is better than only one, as the loss in the graph has a smaller loss value and convergence, although the accuracy is different by only 0.5%, with 85.25% and 85.75%.

Table 5 proves that implementing a freeze layer may decrease the model performance. This may happen because the parameters in the first and second layers may not fit enough to the current model configuration and dataset. In addition, the target dataset's characteristics may also be different from the pre-trained dataset.

5.4 Using fusion RGB + optical flow

We also train the optical flow similarly to the RGB image model configuration. The model with input containing 74 frames was also trained from scratch and using transfer learning. The result of some training schemes can be seen in Table 6 where training from scratch even shows a better result than the RGB image. The use of transfer learning also improves it. The maximum accuracy using optical flow as input is 88.5% with Kinetics pre-trained. It is followed by Kinetics + Moments in Time pre-trained with slightly lower accuracy by 0.5 difference.

After we trained the model with RGB image and optical flow, we tried to fuse the model's output. The best performance of each stream will be added to the output layer and computed as the softmax. The RGB image input configuration only uses model pre-trained on Kinetics + Moments in Time with location range of the sampled frames of 49–149, while the optical flow will use all the trained schemes. The highest configuration uses the fusion of the RGB image input with the optical flow that implements Kinetics pre-trained model, producing the accuracy of 90.5% as shown in Table 7.

This study outperforms some models that use RWF-2000 as a benchmark. As we can see in Table 8, our method reaches high accuracy by fusion the output from the model of RGB image and optical flow.

Mainstream research on deep learning-based video recognition models provides limited information on time complexity and the number of parameters. However, the architecture typically involves a large number of parameters, such as multi-stream convolutional and optical flow networks, along with multiple LSTM layers, which are computationally intensive but not discussed in the research [23]. Video recognition experts in the computer vision field focus on accurate predictions over real-time decision making. The parameters for various

Table 6 Accuracy of Using Pre-trained Model with Input of Optical Flow

Pre-trained	Train	Val
Scratch	100	86
Kinetics	100	88.5
Kinetics + Moments in Time	100	88

Table 7 Accuracy of Fusion Scheme

Scheme of Fusion	Val
RGB (49-149KM) + OF (Scratch)	87
RGB (49-149KM) + OF (K)	90.5
RGB (49-149KM) + OF (KM)	89.75

deep models can be analyzed from Table 9 where all the models are run under Intel Core™ i7-8700 CPU @ 2.90GHz machine, with most models prioritize accuracy. Some models, such as the I3D features-based mechanism by Carreira et al., offer a balance between the number of parameters and accuracy, with 12.3 million parameters for RGB or optical flow input and 24.6 million for a two-stream model. Cheng et al. achieved the best combination of parameters and accuracy, with 87.25% accuracy using only 0.27 million parameters, by using depth-wise separable convolutions from Pseudo-3D Residual Networks and MobileNet to incorporate 3D convolutional layers in their model. However, our proposed two-stream with 3D ResNet as backbones achieves highest accuracy even though the further works remain to make model more efficient (faster and less memory) without sacrificing accuracy. One candidate is performing pruning after training.

5.5 Classification video analysis

Videos in the dataset have different characteristics and various contents. Therefore, it also can test how far the models can understand the video input when doing the classification task. For example, we analyze some videos in RWF-2000 when using our model, Fusion RGB + Optical Flow. The model's success can be determined by how various the input can be. It includes objects' movement, the activity's location, number of objects, and the occurrence of biases in the videos. Some cases also show that one stream may be bad due to the interference of that surveillance camera.

Figure 15 is an example of a video that our model successfully classifies. The video contains a man grabbing a woman inside the lift and pushing her away. The quality of the video also shows gentle movement as the frame changes. On the other hand, the optical flow clearly shows massive movements with the direction of the objects. The RGB stream and optical flow can classify even without the fusion.

Table 8 Comparisons of the Proposed Method and Others on RWF-2000

Method	Accuracy (%)
ConvLSTM [24]	77.00
C3D [26]	82.75
I3D (RGB Only) [4]	85.75
I3D (Flow Only) [4]	75.5
I3D (TwoStream) [4]	81.5
Flow Gated Network [5]	87.25
SAM-GhosNet-ConvLSTM [16]	87.5
SepConvLSTM-M [12]	89.75
Ours (Best Version)	90.5

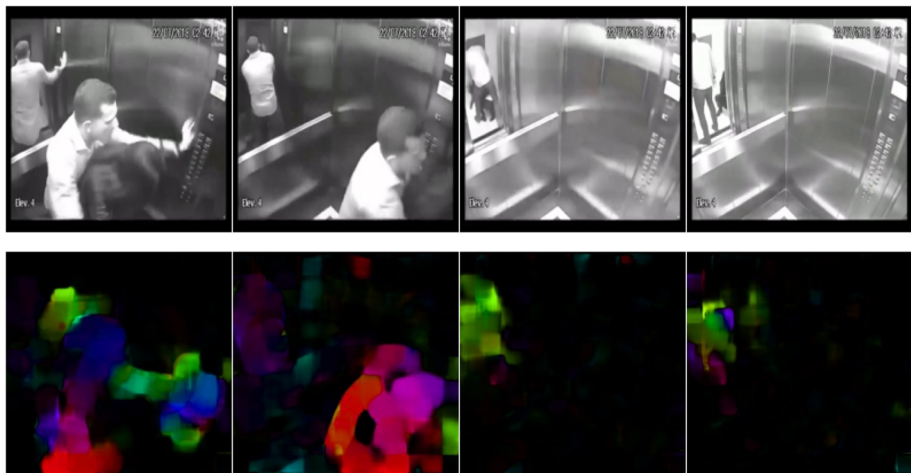
Table 9 Comparisons Speed and Number of Parameters of the Proposed Method and Others on RWF-2000

Algorithm	Params (MB)	FPS	Accuracy
Tran et al. [26]	94.8	0.5	82.75
Sudhakaran et al. [23]	94.8	0.5	77
Carreira et al. [4] (RGB only)	12.3	77.3	85.75
Carreira et al. [4] (OF only)	12.3	77.3	75.75
Carreira et al. [4] (two-stream)	24.6	59.5	81.5
Cheng et al. [16]	0.27	94.7	87.5
Ours	66.6	5	90.5

The RGB stream and two-streams can classify the video in Fig. 16, but not with the optical flow. The video contains chaos in a traditional market. Many objects in that video that recorded by the surveillance camera. The optical flow has too many biases. It can be the shake of the camera movements or the noises. The biases make the optical inaccurate because it has not captured the objects' movement but rather the unimportant pixels' movement.

The optical flow and two-streams can classify the video in Fig. 17, but not with the RGB stream. The video contains two children playing fight together in a living room labeled as a fighting video. While the original video resolution is a portrait, it will be stretched when scaling and resizing and may affect the RGB stream. We can see that the optical flow in Fig. 17 clearly shows the object's movement. However, the background looks static. We can also see that the RGB picture is also stretched, and the color space is grayscale. Although the RGB stream failed to classify, the fusion with the optical flow makes the two-stream successfully do the task because the model's output with softmax activation shows a higher probability of actual label.

Fig. 18 is an example of a video that our model failed to classify. The video contains ground fighting in a place like a basement or a garage. It was recorded from a surveillance camera with low resolution, and frame rate as the RWF-2000 claim has all 30 frames per

**Fig. 15** Good case video where our model can classify

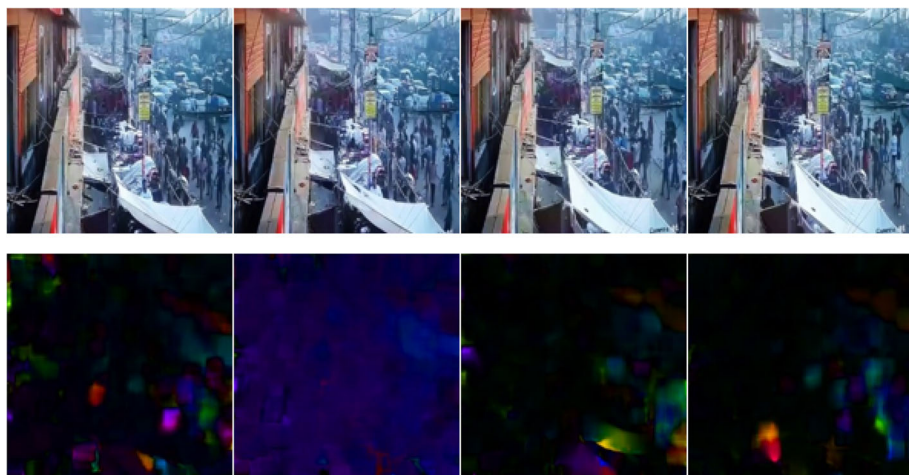


Fig. 16 RGB stream and two-stream can classify this video

second video. If we see this video, it is only like one frame per second because it lacks movement and frame change. The optical flow also does not show significant movement of pixels. There are 6 videos that look like Fig. 18 (a video that separated as 6 parts) in validation dataset.

Another video that fails to be classified may have a lack of movements, too much object movement, camera motion, and the act of violence is too far from the center (due to the validation phase using center spatial crop). Too many noises also affect the quality of optical flow, yielding unimportant information. If one-stream failed to classify a video, it does not mean that two-stream also failed. The fusion by adding the model output using softmax

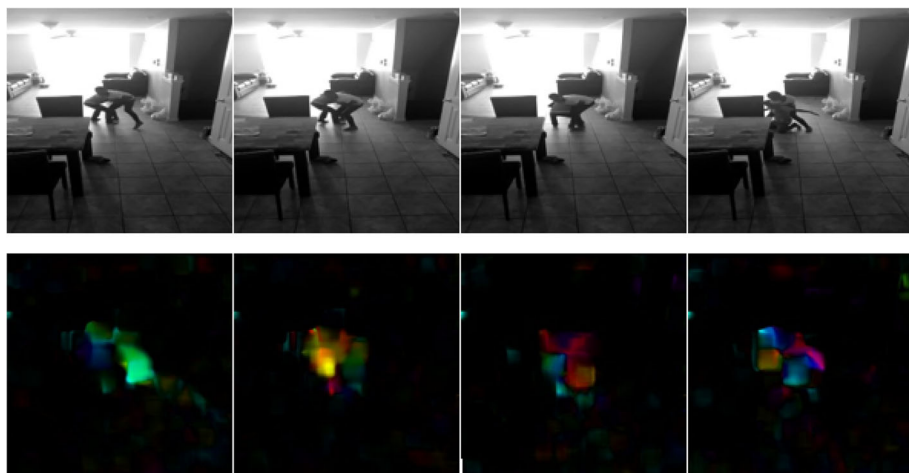


Fig. 17 Optical flow and two-stream can classify this video

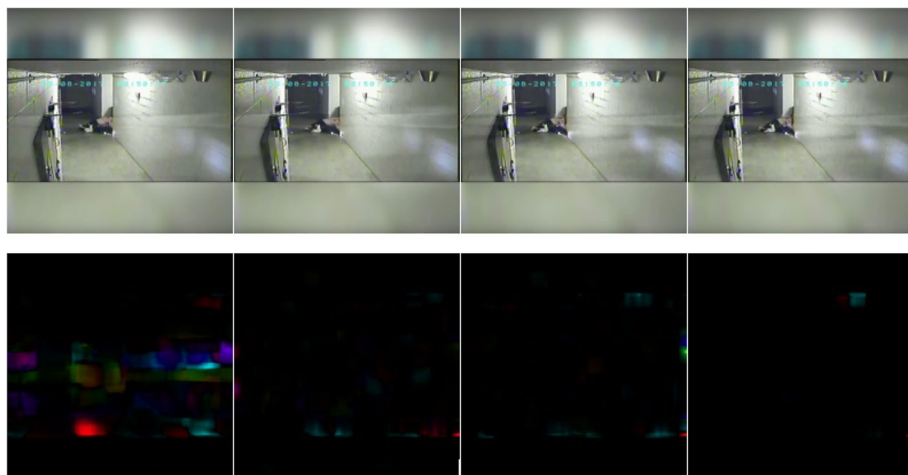


Fig. 18 Bad case video where our model can not classify

activation (two-stream) is complementary and improves rather than RGB stream or optical flow only.

5.6 Discussion

There are some insights that we found in our study. Specifically, data augmentation affects the performance of the model when performing classification task. In this study, some data augmentations focus on the spatiotemporal space in which the images are stacked. It is important to make the sequence of images in an ordered stack because it will affect the information or the knowledge provided for training. Broadly defined, this study uses a random spatiotemporal crop that initiates the starting frame randomly. On the other hand, a video-wise random spatiotemporal crop selects randomly sorted frames in the video. We also proposed a custom spatiotemporal crop that uses a fixed location within specific sampling ranges.

When several frames are utilized, a random spatiotemporal crop performs better as the number of selected frames is increased. A few randomized frames in the video-wise random spatiotemporal crop yield a gap between the frames. It only provides a few and even a piece of cluttered information if a gap is too wide from frame to frame. Therefore, we should include more frames if we conduct a video-wise random spatiotemporal crop. A random spatiotemporal crop is suitable for the dataset with various frames.

Custom spatiotemporal is applicable when the video's length is known and tends to be normalized. For this study, RWF-2000 has videos that have been normalized by cropping for every 5 seconds. We have divided it into three parts: beginning, middle, and end. As shown by Tables 3 and 4, when we involve the middle part, it shows a better result than the other. However, this may have different results with different cases on other datasets.

Another insight of this study is that the more number of frames included will make the model heavier when performing the task. However, fewer frames will give them little information. On the other hand, when the number of frames is too much, it also makes information biased. In this study, using 50–100 frames is experimentally ideal. That amount comes from the $\frac{n}{3}$ of the total n frames.

6 Conclusion

Based on our study, 3D ResNet-18, with some improvement, shows robust performance in video-based violence classification. The number of frames input affects the knowledge stored in the model. The ideal number of frames as input may increase the model's performance. Transfer learning also enhances model accuracy due to the initial parameters. While using transfer learning, the freezing layer on the first and second layers does not always show a better result. The utilization of data augmentation in this study potentially improves performance. Specifically, the augmentation of the spatial and spatiotemporal crops provides various patterns for the model to improve generalization. Our proposed method reached 90.5% by using 3D ResNet-18 with the custom spatiotemporal crop, transfer learning, and fine-tuning of the pre-trained model from Kinetics and Moments in Time dataset using location range of the sampled frames of 49–149 and fusion of RGB and optical flows stream. Our best model's accuracy outperforms some violence detection works that use the RWF-2000 dataset.

Availability of data and materials The datasets generated during and/or analysed during the current study are available in the RWF-2000 repository, <https://github.com/mchengny/RWF2000-Video-Database-for-Violence-Detection>

Declarations

Conflicts of interests/Competing interests The authors have no conflicts of interest to declare that are relevant to the content of this article

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